

scRareRefine: risk-controlled rescue of low-frequency cell types under rare-label scarcity

[Author names]

[Affiliations]

Correspondence: [email]

Manuscript category: Gene expression

Article type: Original Paper

Running title: Risk-controlled rare-cell rescue

Structured Abstract

Motivation: Semi-supervised single-cell annotation models can miss low-frequency cell types when only a small number of rare-class labels are available. A useful refinement method should recover these missed cells without creating a broad false rare-cell call set.

Results: We introduce scRareRefine, a post-hoc refinement module for scANVI that combines train-derived latent prototypes with validation-only conformal calibration. scRareRefine targets a pre-specified low-frequency class and abstains when validation evidence does not support rescue. Across eight human and mouse scRNA-seq datasets, nine methods, four rare-label budgets and three seeds, the scarce-label settings contained only 0.0206%–0.3526% labeled rare cells in training. In these settings, scRareRefine achieved mean rare-cell F1 of 0.801, compared with 0.646 for scANVI and 0.665 for kNN, and was best or tied-best in 23 of 24 dataset-budget cells. Its maximum false rare-call rate was 0.009878 under a pre-specified 0.01 budget.

Availability and Implementation: Source code, configuration files, and result-generation scripts will be released at [GitHub URL] and archived at [Zenodo DOI]. The implementation is in Python using scvi-tools, Scanpy and AnnData.

Contact: [email]

Supplementary information: Supplementary data are available at [URL].

1 Introduction

Single-cell RNA sequencing has made it routine to profile complex tissues at cellular resolution, but the cell types that are most biologically interesting are often not the most abundant. Rare immune subsets, endocrine subtypes, tissue-resident stromal cells and lineage intermediates may account for only a few percent of a dataset. They are also the classes for which labels are most fragile: small reference panels contain few positive examples, batch-heldout evaluation can place the target class in different technical contexts, and semi-supervised classifiers may collapse an under-represented class into a nearby majority class.

This creates a practical failure mode for cell-type annotation. A model can achieve acceptable global accuracy while systematically losing the cell type under study. The problem is not simply that the class is rare in the full dataset. For a supervised or semi-supervised annotation model, the more direct quantity is the fraction of training cells that are labeled examples of the target rare class. In the experiments below, every

formal target rare class has dataset-level prevalence below 5%, but the scarce-label regimes are more severe: after downsampling rare labels in the training split, labeled rare cells comprise only 0.0206%–0.3526% of all training cells in the main scarce-label settings.

Existing annotation methods approach the problem from different directions. scANVI learns a semi-supervised latent representation and classifier from labeled and unlabeled cells [1, 2]. CellTypist, TOSICA and related supervised systems provide scalable annotation using marker-informed or transformer-based models [3, 4]. Rebalancing and rare-cell detection methods, including scBalance and scCAD, are designed to counter class imbalance or discover unusual populations. Prototype-based approaches such as ProtoCloud use geometric summaries of reference labels. These methods are valuable, but they do not directly address a specific deployment question: when a known low-frequency target class is missed by a semi-supervised classifier, can we selectively recover likely false negatives while controlling false rare-cell calls?

scRareRefine is designed for this setting. It is not a de novo rare-cell discovery method and it does not change the scANVI training objective. Instead, it is an inductive post-hoc module applied after scANVI. It builds latent-space prototypes from labeled training cells, scores candidate cells by their proximity to the target rare prototype relative to other class prototypes, and uses validation cells to decide whether rescue is needed and how permissive the rescue rule should be. The final relabeling threshold is calibrated from validation non-rare cells under a fixed false rare-call budget. If the target class is not geometrically separable, if the baseline already recovers the validation rare cells, or if no candidate rank satisfies the validation safety criterion, the method abstains and returns the baseline predictions.

We evaluate scRareRefine against eight baselines or alternative methods across eight datasets covering human and mouse samples, multiple tissues, three random seeds and four rare-label budgets. The benchmark includes six human datasets and two mouse Tabula Muris Senis 10x add-on datasets, enabling cross-tissue and cross-species breadth while preserving within-dataset train-validation-test evaluation. The main claim is that scRareRefine recovers rare-cell annotations under rare-label scarcity with a pre-specified false rare-call budget, and that it abstains when rescue is not warranted.

2 Materials and Methods

2.1 Problem setup

Let a single-cell dataset contain gene-expression matrix X , cell-type labels y for a subset of cells, and a pre-specified target rare class r . The target class is supplied by the dataset configuration rather than inferred automatically from a frequency threshold. We use the full labeled dataset only to define the evaluation target and then construct inductive train, validation and test splits. Model fitting, prototype estimation, rank selection and threshold calibration use only training and validation information. Test labels are reserved for final evaluation.

We distinguish two notions of rarity. The first is the dataset-level target prevalence,

$$\pi_r = N_r/N,$$

where N_r is the number of cells of the configured target rare class and N is the dataset size. This is the biological and benchmarking definition used to select target classes. In the formal benchmark, after excluding the exploratory `tabula_lung_stroma` dataset, all configured target rare classes have $\pi_r < 5\%$.

The second is rare-label availability in the training split,

$$\lambda_r = L_{r,\text{train}}/N_{\text{train}},$$

where $L_{r,\text{train}}$ is the number of labeled rare-class training cells retained after rare-label downsampling and N_{train} is the total number of training cells. This is the operational scarcity seen by the classifier. The label

budgets `rare_train_size = 0.01, 0.05, 0.10` retain approximately 1%, 5% or 10% of training rare labels, with a minimum of five labeled rare cells when available; `rare_train_size = all` retains all training rare labels. Across the formal eight-dataset benchmark and three seeds, the scarce-label settings correspond to $\lambda_r = 0.0206\%$ – 0.3526% of training cells.

2.2 Data splitting and label masking

The primary experiments use batch-heldout splits. Cells are partitioned into training, validation and test sets by batch when dataset metadata supports this design, with fallback logic handled by the preprocessing pipeline. Because batches have unequal sizes, the resulting split proportions need not exactly match nominal ratios, but the split is fixed before model fitting and is recorded in the run manifest. This design tests whether a rare-cell recovery rule transfers across held-out experimental batches rather than only across randomly sampled cells.

Within the training split, rare-class labels are downsampled according to the selected label budget. Non-rare labeled cells remain available, while rare cells whose labels are masked are treated as unlabeled for semi-supervised training. Validation and test labels are not downsampled for evaluation; validation labels are used only for method selection and calibration, and test labels are used only for final metrics.

2.3 scANVI backbone and prototype scoring

The backbone model is scANVI, trained through scvi-tools on highly variable genes selected within the pipeline. The model first learns a latent representation with scVI and then fits the semi-supervised scANVI classifier using labeled and unlabeled training cells. For each split, the pipeline exports predicted labels, class probabilities and latent embeddings. These exported embeddings are shared by scRareRefine and by comparison scripts where applicable, with manifest checks to prevent accidental reuse across incompatible configurations, splits or datasets.

scRareRefine builds class prototypes in the scANVI latent space using labeled training cells. For each class c , the prototype μ_c is the mean latent vector of labeled training cells of class c . A class-specific radius ρ_c is estimated from within-class distances to the prototype, using the median distance with a conservative fallback for extremely small classes. Distances from a cell embedding z_i to each prototype are normalized by these radii.

The rare-membership score is computed from normalized prototype distances so that cells closer to the target rare prototype, relative to competing prototypes, receive higher scores. We also compute the rank of the rare prototype among all class prototypes for each cell. A cell is considered a rescue candidate only if the scANVI prediction is not the target rare class and the rare prototype rank is at most k , where k is selected on validation data from the grid $\{1, 2, 3\}$. This candidate rule limits rescue to cells that are already geometrically close to the rare prototype.

2.4 Validation-only gates and conformal calibration

scRareRefine applies three validation-only safeguards before relabeling test cells. First, a separability gate compares the distance between the target rare prototype and the nearest non-rare prototype against the within-rare radius. If the target rare class is insufficiently separated, with separability below 1.3, the method abstains. This constant is treated as a pre-specified conservative prior rather than tuned per dataset.

Second, a necessity gate checks whether rescue is supported by validation errors. If the validation split contains too few missed rare cells by the baseline, or if scANVI already recovers the target rare class on validation, scRareRefine abstains. This prevents unnecessary relabeling when the baseline is already saturated or when validation evidence is too sparse to select a reliable rescue rule.

Table 1: Formal benchmark datasets and target rare classes.

Configuration	Context	Target rare class
immune_dc	human immune atlas	ASDC
pancreas_baron	human pancreas	gamma / epsilon
pancreas_integrated	integrated human pancreas	endothelial
tabula_lung_endo	human lung	endothelial subtype
tabula_sapiens_stomach	human stomach	mast cell
tabula_small_intestine	human small intestine	intestinal tuft cell
mouse_lung_tms_10x	mouse lung, TMS 10x	vein endothelial cell
mouse_pancreas_tms_10x	mouse pancreas, TMS 10x	pancreatic D cell

Third, the method selects the maximum candidate rank k from $\{1, 2, 3\}$ using validation performance under a safety constraint. Candidate ranks are evaluated by validation rare F1 subject to a Wilson upper-bound constraint on false rare-cell calls. Ties are broken toward the smaller rank, yielding a more conservative rule.

After a rank is selected, the final rare-score threshold is calibrated from validation non-rare cells. Let s_i be the rare-membership score for validation cells whose true label is not r . The threshold is a finite-sample high quantile corresponding to a pre-specified false rare-call budget $\alpha = 0.01$ [5, 6]. At test time, a non-rare scANVI prediction is changed to the target rare class only if the cell passes the selected rank rule and its rare-membership score exceeds the calibrated threshold. Otherwise, the original scANVI prediction is retained.

2.5 Datasets, methods and metrics

The formal benchmark contains eight datasets. Six are human datasets spanning immune, pancreas, lung, stomach and small intestine contexts. Two mouse Tabula Muris Senis 10x datasets were added to broaden the evaluation across species and tissues. The configured target rare classes are fixed in dataset configuration files and are not selected post hoc from model performance.

We compare scRareRefine with scANVI, kNN on the shared latent representation, CellTypist, scBalance, ProtoCloud, HiCat, scCAD and TOSICA. All methods are evaluated on the same dataset splits, seeds and rare-label budgets where applicable. The complete comparison grid contains eight datasets, four label budgets, three seeds and nine methods, for 864 runs. The final result table contains 864 successful runs.

The primary metric is rare-class F1 on the independent test split. We also report rare-class precision and recall, because the method is intended to recover rare false negatives rather than only improve aggregate annotation accuracy. The safety metric is the false rare-call rate among true non-rare cells, reported in the result files as `rare_fp_rate`. The scarce-label region is defined as `rare_train_size` in $\{0.01, 0.05, 0.10\}$.

3 Results

3.1 Rare-label scarcity is more severe than target-class prevalence

All formal target classes are low-frequency classes in their respective datasets, with full-dataset prevalence below 5%. However, the label-scarcity experiment is governed by the number of labeled rare cells available to the training process. In the most stringent settings, the classifier sees only a handful of rare labels against thousands of non-rare training cells. For example, in `immune_dc` at `rare_train_size = 0.01`, the training split contains 277 rare cells and 11,014 non-rare cells, but only five rare cells remain labeled, giving $\lambda_r = 5/(277 + 11014) = 0.0443\%$.

Table 2: Mean performance in scarce-label settings. Max false rare-call rate is the maximum over scarce-label runs.

Method	Rare F1	Recall	Mean FP rate	Max FP rate
scRareRefine	0.8008	0.7543	0.0012	0.0098
kNN	0.6645	0.5883	0.0002	0.0035
scANVI	0.6463	0.5855	0.0003	0.0023
scBalance	0.5827	0.4967	0.0001	0.0021
ProtoCloud	0.5592	0.4776	0.0002	0.0018
CellTypist	0.5514	0.4688	0.0002	0.0012
scCAD	0.4726	0.6698	0.0246	0.0752
TOSICA	0.4345	0.3918	0.0030	0.0209
HiCat	0.1445	0.1193	0.0000	0.0002

Table 3: Paired scarce-label rare-F1 comparisons. Differences are paired by dataset, rare-label budget and seed.

Comparator	Win	Tie	Loss	Mean Δ F1	95% CI
scANVI	41	30	1	+0.1545	[+0.0956,+0.2199]
kNN	59	8	5	+0.1363	[+0.0967,+0.1773]
CellTypist	65	4	3	+0.2494	[+0.1950,+0.3049]
scBalance	61	5	6	+0.2181	[+0.1642,+0.2737]
ProtoCloud	62	6	4	+0.2416	[+0.1884,+0.2972]
HiCat	66	5	1	+0.6563	[+0.5795,+0.7291]
scCAD	66	5	1	+0.3282	[+0.2815,+0.3771]
TOSICA	67	2	3	+0.3663	[+0.3064,+0.4261]

This framing clarifies the evaluation target. The task is not to discover an arbitrary cluster below a fixed frequency threshold. The task is to preserve and recover a known low-frequency class when labeled positive examples are scarce enough that a semi-supervised classifier may erase the class from its predictions.

3.2 scRareRefine improves rare-cell F1 in scarce-label settings

In the scarce-label region, scRareRefine achieved the highest mean rare-cell F1 across methods. Averaged over datasets, seeds and scarce-label budgets, its rare-cell F1 was 0.8008, compared with 0.6645 for kNN and 0.6463 for scANVI. The improvement is driven by recall recovery: scRareRefine mean rare recall was 0.7543, compared with 0.5883 for kNN and 0.5855 for scANVI.

The method was also robust at the dataset-budget level. Collapsing over seeds gives 24 scarce-label dataset-budget cells. scRareRefine was exactly best or tied-best in 23 of 24 cells. In the only non-best cell, `tabula_small_intestine` at `rare_train_size` = 0.10, ProtoCloud achieved rare F1 of 0.9848 and scRareRefine achieved 0.9823, a difference of 0.0025.

3.3 Paired comparisons show consistent gains

We compared scRareRefine against each method in matched dataset-budget-seed units in the scarce-label region ($n = 72$ paired units per baseline). Against scANVI, scRareRefine had 41 wins, 30 ties and one loss, with mean paired Δ F1 of +0.1545 and bootstrap confidence interval [+0.0956,+0.2199] (one-sided paired Wilcoxon $P = 8.82 \times 10^{-9}$). The large number of ties is expected: if validation evidence indicates that rescue is unnecessary or unsafe, scRareRefine returns the baseline prediction.

Because paired units are correlated across seeds and some rare-label budgets collapse to the same effective label count, these P values should be read as directional evidence alongside effect sizes and confidence

intervals, not as strictly independent-sample tests.

3.4 False rare-cell calls remain within the pre-specified budget

Recovering more rare cells is only useful if the rare label remains specific. Across the complete eight-dataset comparison, including the `all` label-budget setting, the maximum false rare-call rate observed for `scRareRefine` was 0.009878, below the pre-specified $\alpha = 0.01$ budget. This maximum occurred in `mouse_lung_tms_10x` at `rare_train_size = all` for one seed. In the scarce-label region, the largest values remained close to, but below, the same budget.

The comparison methods show different safety profiles. Some methods, such as `kNN` and `scANVI`, have low false rare-call rates but miss many rare cells. Others, such as `scCAD` in this benchmark, recover more rare cells but can exceed the false rare-call budget. `scRareRefine` occupies the intended middle ground: it increases rare recall relative to `scANVI` while using validation calibration to constrain broad rare-label expansion.

3.5 Cross-tissue and mouse add-on datasets support broader applicability

The original human benchmark covers multiple tissue contexts, including immune, pancreas, lung, stomach and small intestine datasets. The two mouse Tabula Muris Senis 10x add-on datasets extend this breadth to a second species and additional tissue contexts while keeping the same evaluation contract. These mouse datasets are not treated as a separate domain-transfer experiment in which a model is trained on human cells and tested on mouse cells. Rather, they test whether the same label-scarcity and rescue procedure remains useful outside the original human-only panel.

The inclusion of mouse lung and mouse pancreas reduces the risk that the method is tuned to a single organism or tissue geometry. The main aggregate results include these two datasets, and the complete 864-run comparison succeeded without method-specific changes to the `scRareRefine` algorithm.

4 Discussion

The central result is that a constrained post-hoc rescue module can recover rare-cell annotations lost by a semi-supervised backbone under severe label scarcity. `scRareRefine` improves rare-cell F1 and recall across an eight-dataset benchmark while keeping the observed false rare-call rate below a fixed 1% budget. The method is deliberately narrow: it targets one configured low-frequency class, builds prototypes only from labeled training cells, calibrates thresholds only on validation cells, and abstains when validation evidence is insufficient.

This narrowness is a practical advantage. Many single-cell studies do not need a fully new annotation model; they need to know whether a small, known cell type has been erased by the combination of class imbalance and batch shift. Because `scRareRefine` operates after `scANVI`, it can be attached to an established semi-supervised workflow and audited through its gates, candidate ranks and calibrated threshold. When it changes predictions, the change is traceable to latent prototype proximity and validation-calibrated rare-score evidence. When it does not change predictions, the abstention is also informative.

The evaluation highlights why rare-cell benchmarking should report both target prevalence and labeled rare-cell availability. A target class below 5% is biologically low-frequency, but the model’s difficulty depends on how many labeled rare examples remain in the training set. Reporting λ_r makes clear that the method is being tested under a true rare-label regime, rather than simply on datasets containing low-frequency classes.

There are several limitations. First, `scRareRefine` assumes the target rare class is known in advance. It is therefore complementary to *de novo* rare-cell discovery methods rather than a replacement for them.

Second, the current implementation refines a single target rare class per run. Multi-rare-class settings may require class-wise calibration or a joint error-budget allocation. Third, the method inherits the quality of the scANVI latent representation. If rare and majority cells are not separable in the latent space, the conservative gates will abstain or recall will remain bounded. Fourth, while the benchmark now includes mouse add-on datasets, it does not test direct human-to-mouse model transfer. Finally, the current manuscript reports computational results from public scRNA-seq datasets; experimental validation of newly recovered rare cells would require independent biological assays or marker-level confirmation.

5 Conclusion

scRareRefine provides a validation-calibrated post-hoc rescue procedure for known low-frequency cell types in scRNA-seq annotation. By combining train-only latent prototypes, validation-selected candidate ranks, conformal score thresholds and abstention gates, it improves rare-cell recovery under severe rare-label scarcity while maintaining a fixed false rare-call budget. The current eight-dataset benchmark supports scRareRefine as a practical refinement module for studies where preserving a known rare cell type is more important than maximizing aggregate annotation accuracy.

Data availability

All datasets used in this study are derived from public single-cell resources, including human atlas datasets and CELLxGENE/Tabula Muris Senis mouse 10x subsets. Processed dataset identifiers, download instructions and preprocessing scripts will be provided in the repository and Supplementary Data. No new sequencing data were generated for this study.

Code availability

The scRareRefine implementation, configuration files, comparison scripts, plotting scripts and result manifests will be made freely available at [GitHub URL] under [license]. The submitted version will be archived at [Zenodo DOI]. Reproducibility instructions will include the two conda environments used for the comparison grid: `scanvi311` for the scANVI/scRareRefine pipeline and most baselines, and `sandbox310` for baselines requiring older dependency stacks.

Funding

This work was supported by [funding information].

Conflict of Interest

The authors declare no competing interests.

Acknowledgements

The authors thank the maintainers of scvi-tools, Scanpy, AnnData, CELLxGENE, Tabula Sapiens and Tabula Muris Senis for public data and software resources.

Internal author notes, not for submission

This is an AI-assisted working draft for author rewriting. Bioinformatics follows the ISCB policy on large language models; before submission, the manuscript must be substantially revised by the authors, all references and numerical claims must be checked against final artifacts, and AI assistance must be disclosed where required.

Open items before submission: verify every reference and DOI; decide whether ablation results should be rerun on all eight datasets or clearly labeled as six-human ablation; add final GitHub and Zenodo URLs; convert to the official Bioinformatics or BMC Bioinformatics template after the scientific content is frozen.

References

- [1] Lopez, R., Regier, J., Cole, M.B., Jordan, M.I. and Yosef, N. (2018) Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15, 1053–1058.
- [2] Xu, C., Lopez, R., Mehlman, E., Regier, J., Jordan, M.I. and Yosef, N. (2021) Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models. *Molecular Systems Biology*, 17, e9620.
- [3] Dominguez Conde, C., Xu, C., Jarvis, L.B. et al. (2022) Cross-tissue immune cell analysis reveals tissue-specific features in humans. *Science*, 376, eabl5197.
- [4] Chen, J., Xu, H., Tao, W., Chen, Z., Zhao, Y. and Han, J.-D.J. (2023) Transformer for one stop interpretable cell type annotation. *Nature Communications*, 14, 223.
- [5] Vovk, V., Gammerman, A. and Shafer, G. (2005) *Algorithmic Learning in a Random World*. Springer.
- [6] Angelopoulos, A.N. and Bates, S. (2023) Conformal prediction: a gentle introduction. *Foundations and Trends in Machine Learning*, 16, 494–591.
- [7] The Tabula Sapiens Consortium. (2022) The Tabula Sapiens: a multiple-organ, single-cell transcriptomic atlas of humans. *Science*, 376, eabl4896.
- [8] Tabula Muris Consortium. (2018) Single-cell transcriptomics of 20 mouse organs creates a Tabula Muris. *Nature*, 562, 367–372.
- [9] Tabula Muris Senis Consortium. (2020) A single-cell transcriptomic atlas characterizes ageing tissues in the mouse. *Nature*, 583, 590–595.
- [10] Baron, M., Veres, A., Wolock, S.L. et al. (2016) A single-cell transcriptomic map of the human and mouse pancreas reveals inter- and intra-cell population structure. *Cell Systems*, 3, 346–360.

